

# Speed, Distance and Time — Paper A

MEDIUM • 40 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer every question in the space provided. Show your working — method marks are awarded even when the final answer is wrong.
- A calculator is allowed, but every number on this paper is designed to work without one.
- Every answer must carry a unit. An answer with no unit does not score the final mark.
- The mark for each question is shown in square brackets on the right.
- Reminder: speed = distance  $\div$  time, and to convert m/s to km/h multiply by 3.6.

## SECTION 1 – THE FORMULA

- 1.** Write down the formula for speed, and state the SI unit of speed. [2]

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- 2.** Calculate the speed in each case.

- (a)** 100 m travelled in 20 s. [1]

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- (b)** 240 km travelled in 3 hours. [1]

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- (c)** 45 m travelled in 6 s. [2]

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- 3.** Calculate the distance travelled.

- (a)** 12 m/s for 30 s. [1]

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**(b)** 60 km/h for 2 hours. [1]

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**4.** Calculate the time taken.

**(a)** 400 m at 8 m/s. [1]

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**(b)** 150 km at 50 km/h. [1]

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SECTION 2 – UNITS

**5.** Convert each speed.

**(a)** 20 m/s into km/h [1]

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**(b)** 90 km/h into m/s [1]

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**(c)** 5 m/s into km/h [1]

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**6.** Convert each time.

**(a)** 30 minutes into hours [1]

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**(b)** 2 hours 15 minutes into hours [1]

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**(c)** 3 minutes into seconds [1]

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**7.** A walker's journey is recorded below.

[3]

|               |   |    |    |    |
|---------------|---|----|----|----|
| Time (s):     | 0 | 10 | 20 | 30 |
| Distance (m): | 0 | 15 | 30 | 45 |

Calculate the walker's speed, and state how you can tell it is constant.

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**8.** State what a horizontal line on a distance-time graph tells you about the object, and explain why.

[2]

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**9.** A cyclist travels 6 km in 15 minutes. Calculate her speed in km/h.

[3]

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**1** A student calculates that his walking speed is 50 m/s. Explain why this answer must **0.** be wrong.

[2]

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**1** A car travels at 15 m/s for 40 s. Calculate how far it goes, giving your answer in **1.** metres and in kilometres.

[2]

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**1** State the difference between average speed and instantaneous speed.

[2]

**2.**

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*End of paper. Check every answer has a unit or a form the question actually asked for.*